

Simmaco Di Lillo

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Education

Ph.D. Candidate in Mathematics Sep 2023 – Present

University of Rome “Tor Vergata”

Thesis: “The Geometry of Random Neural Networks”

Supervisor: Prof. Domenico Marinucci

Master in Mathematics (Applied Mathematics curriculum) Sep 2021 – Sep 2023

University of Rome “La Sapienza”

Thesis: *Stein’s method and applications to change point detection*

Supervisor: Prof. Alessandra Faggionato

Final grade: 110/110 with honors

Bachelor in Mathematics (Applied Mathematics curriculum) Sep 2018 – Sep 2021

University of Pisa

Thesis: *The SIR model: bottom-up on a network*

Supervisor: Dr. Fabio Durastante

Final grade: 110/110 with honors

Research Visits

Ph.D. Visiting Student Feb 2025 – Mar 2025

Center for Computational Mathematics, Flatiron Institute, Simons Foundation

Host supervisor: Dr. Alberto Bietti

Ph.D. Visiting Student Jul 2026 – Aug 2026

Department of Statistical Science, University College London

Host supervisor: Dr. Alessandra Cipriani

Publications

- (2026) S. Di Lillo, L. Maini, D. Marinucci. *Phase Transitions in the Fluctuations of Functionals of Random Neural Networks*.
arXiv: [2604.19738](#)
- (2026) S. Di Lillo, C. Macci, B. Pacchiarotti. *Large deviation principles and functional limit theorems in the deep limit of wide random neural networks*.
arXiv: [2601.04677](#)
- (2025) S. Di Lillo. *Critical Points of Random Neural Networks*.
arXiv: [2505.17000](#) (to appear in *Springer INdAM Series*)
- (2025) S. Di Lillo, D. Marinucci, M. Salvi, S. Vigogna. *Fractal and Regular Geometry of Deep Neural Networks*.
arXiv: [2504.06250](#) (to appear in *The Annals of Applied Probability*).
- (2025) S. Di Lillo, D. Marinucci, M. Salvi, S. Vigogna. *Spectral complexity of deep neural networks*. *SIAM Journal on Mathematics of Data Science*, **7**(3). doi: [10.1137/24M1675746](#).
- (2025) S. G. Grisanti, et al. *Neurological long COVID in the outpatient clinic: Is it so long?* *European Journal of Neurology* **32** (2025). doi: [10.1111/ene.16510](#).
- (2023) L. Pezza, S. Di Lillo. *Fractional Dynamical Systems solved by a Collocation Method based on Refinable Spaces*. *Axioms* **12**(5). doi: [10.3390/axioms12050451](#).

Co-supervision and Refereeing Activity

- Referee for *Electronic Journal of Probability*, *Bernoulli*, *Applied and Computational Harmonic Analysis*, *Journal of Machine Learning Research*.
- Valerio Bretti, Master Thesis in Mathematics: *Orthogonal polynomials and applications to Neural Networks*, academic year 2023/24.
- Agnese Profico, Bachelor Thesis in Mathematics: *Spazi di funzioni e Reti Neurali: il teorema di Bach e Bietti*, academic year 2023/24.

Selected Talks and Posters

1. *The Geometry of Neural Networks* 16 Jul 2026
University College London (invited speaker)
2. *Spectral and Geometric Phase Transitions in Deep Neural Networks* 11 Jun 2026
The 5th Italian Meeting on Probability and Mathematical Statistics. University of Palermo (contributed talk)
3. *Three Regimes of Spectral and Geometric Complexity in Deep Networks* 21 Jan 2026
4th Workshop of UMI Group “Math4AiML” University of Rome “La Sapienza”. (contributed talk)
4. *The Geometry of Random Neural Networks* 28 Nov 2025
Statistics Seminar. Collegio Carlo Alberto. Torino (invited speaker)
5. *The Geometry of Random Neural Networks* 8 – 12 Sep 2025
Summer School: Mathematical methods for high-dimensional data. University of Rome “La Sapienza”. (poster)
6. *A Geometrical Overview of Random Neural Networks* 16 – 20 June 2025
PRIN GRAFIA Conference, Cortona. (invited speaker)
7. *On the Fractal Geometry of Random Neural Networks.* 7 – 9 May 2025
Fractional Calculus, Probability and Non-Local Operators 2025. University “Luigi Vanvitelli”. (poster)
8. *Random Neural Networks: Spectral Complexity. Fractal or Regular Boundary?* 20 March 2025
University of Trento. (invited speaker)
9. *On the Spectral Complexity of Deep Neural Networks.* 29 – 31 Jan 2025
3rd Workshop of UMI Group “Math4AiML” University of Bari. (contributed talk)
10. *Spectral Complexity of Deep Neural Networks.* 6 – 15 Aug 2024
Machine Learning Theory Summer School, August 2024, Princeton University. (poster)

Selected Schools

- Summer School: *Mathematical methods for high-dimensional data* 8 – 12 Sep 2025
University of Rome “La Sapienza”
- PhD School: “Network and Risk Propagation” 18 – 31 Jan 2025
Collegio Internazionale Ca’ Foscari. University of Venice.
- Summer School: *High-Dimensional Approximation* 23 – 27 Sep 2024
CIME, Cetraro.
- Summer School: *Machine Learning Theory* 6 – 15 Aug 2024
Princeton University

Work Experience

Co-Founder and CTO

Mar 2025 – Present

Neuture.ai: EduTech startup. Platform helping students navigate European universities and scholarship opportunities.

University Tutor

Mar 2024 – Jun 2025

University of Rome “Tor Vergata”

- Delivered classroom lectures for *Calculus*. Supervisor: Prof. Mario Abundo.
- Delivered classroom lectures for *Geometry*. Supervisor: Prof. Filippo Viviani.
- Assisted in conducting examinations.

Trainee (Multimodal Biomedical Data Analysis)

Nov 2022 – Sep 2023

Policlinico “San Martino” Hospital (Scholarship), Genoa, Italy

Supervisor: Prof. Michele Piana.

- Applied unsupervised AI and statistical models to interpret clinical data associated with long COVID-19 patients.
- Employed Machine Learning techniques to deconvolve MRI data of glioma.
- Used R and Python for data analysis.

University Tutor

Aug 2021 – Nov 2022

University of Rome “La Sapienza”

- Remedial classes for *General Mathematics* (75h). Supervisor: Prof. Eugenio Montefusco.
- Delivered lectures for *Numerical Analysis* (150h). Supervisor: Prof. Laura Pezza.
- Remedial classes for *Calculus and Biostatistics* (75h). Supervisor: Prof. Gianluca Panati.
- Assisted in conducting examinations.

Skills

Programming: C, C++, R, Python, MATLAB, Wolfram (Mathematica), Bash, OCaml, Julia.

Markup: L^AT_EX, HTML.

Other IT: Office suite, Unix-like systems.

Languages: Italian (Native), English (Fluent).

Participation in Research Groups and Centers

Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni

Istituto Nazionale di Alta Matematica “Francesco Severi”

Oct 2023 – Present **Rome Centre on**

Mathematics for Modelling and Data Sciences

Sep 2023 – Present

Department of Mathematics, University of Rome “Tor Vergata”

Awards & Certificates

- 2026: First-Class Diploma of Merit — Italian Civil Protection Department, Presidency of the Council of Ministers
- 2023: Certificate of merit “Flood in Emilia Romagna”, Confederazione Nazionale delle Misericordie d'Italia.
- 2022: Honors program, University of Rome “La Sapienza”.
- 2021: University merit scholarship, University of Rome “La Sapienza”.
- 2020: Base operator of Civil Protection certificate, Local Health Department (Tuscany).
- 2020: Certificate of merit “Pandemic Covid-19”, Confederazione Nazionale delle Misericordie d'Italia.
- 2018: BLSD Certificate, Local Health Emergency Department.

Updated: July 15, 2026